

Deer's Rock: A Persistent, Event-Driven Healthcare Simulation Platform

1. Overview

Healthcare simulation is essential for policy analysis, AI benchmarking, and health information system validation. Existing platforms such as SimPy, AnyLogic, and MedModel support discrete-event simulation but lack three properties that we argue are essential for modern healthcare experimentation: persistent deterministic replay, modular composability of clinical departments, and agent-native experimentation within a single open-source framework.

Deer's Rock is a persistent, event-driven hospital simulation environment built as a reference implementation of the Healthcare Operating Environment (HOE) architecture. The platform models a full Tier A referral hospital in Makassar, Eastern Indonesia, with 9 specialized departments, 50 ICD-10 diagnoses, rule-based clinical agents (doctor, nurse, pharmacist), a 22-drug formulary, and a stochastic disaster scenario engine covering 7 event types. Unlike existing simulators, the calendar engine generates culturally-contextualized patient influx — Lebaran burn injuries, Ramadan fasting-related hypoglycemia — modelled from real Indonesian epidemiology.

The platform is designed around two principles: **simulation as self-critique**, where reports expose modeling assumptions rather than merely summarizing outputs; and **counterfactuals by construction**, where deterministic seeding and snapshot persistence enable controlled branching experiments.

2. Problem and Background

2.1 Existing simulation platforms

General-purpose simulation frameworks such as SimPy [1] and AnyLogic [2] have been widely applied to healthcare modelling. SimPy provides process-based discrete-event simulation in Python but lacks built-in support for deterministic replay, modular domain handlers, or healthcare-specific data models. AnyLogic offers multi-method simulation and has been applied to emergency department crowding [3] and operating room scheduling [4], but is proprietary and does not provide an event-sourced journal for reproducible replay. MedModel [5] is purpose-built for healthcare but is also proprietary. Synthea [6] generates synthetic patient records for health IT testing but does not simulate hospital operations or support agent-based experimentation. None of these platforms combine deterministic replay, culturally-contextualized patient generation, modular departmental composition, and FHIR-compliant interoperability in a single open-source framework.

2.2 Healthcare ML and digital twin platforms

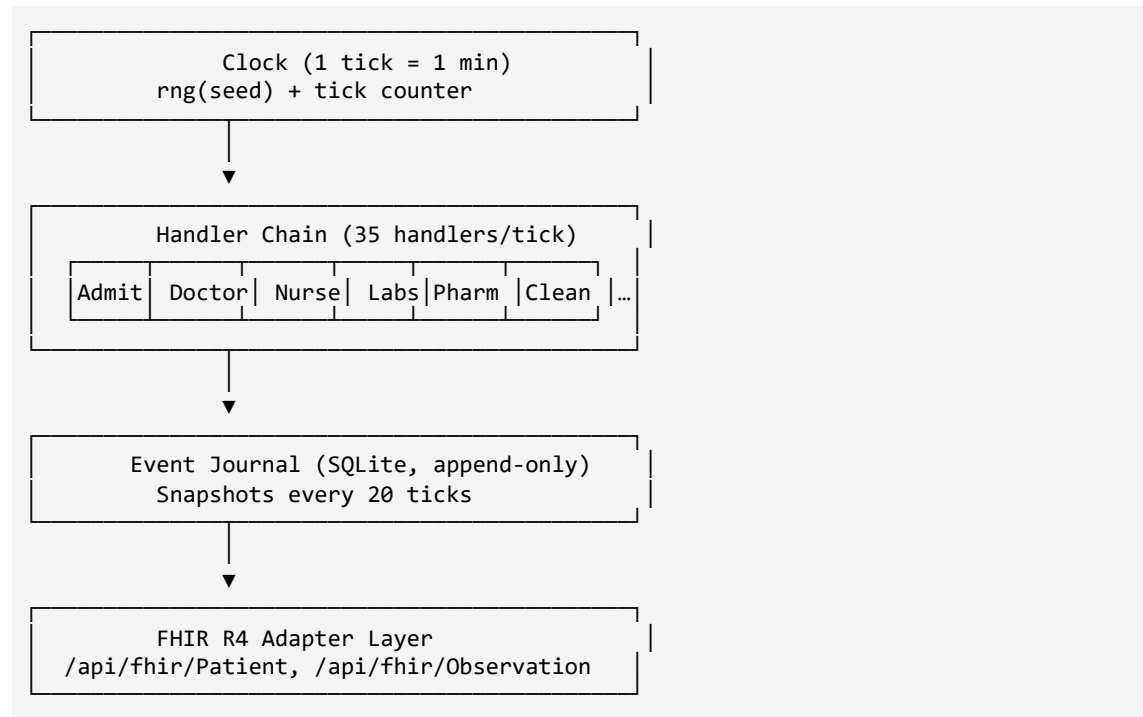
Komorowski et al. [7] demonstrated reinforcement learning for sepsis treatment optimisation using retrospective ICU data, and Gottesman et al. [8] established guidelines for RL in

healthcare. These approaches focus on isolated clinical tasks rather than full hospital operations. Healthcare digital twin initiatives such as the NHS DIGIT programme [9] and Siemens Healthineers [10] are organization-specific, tied to particular hospital data, and generally not open-source.

3. Software Description

3.1 Architecture

Deer's Rock is built on three architectural foundations: a deterministic tick engine, an append-only SQLite event journal, and a modular handler chain.



Tick engine. The simulation advances at 1 tick per second real-time, where each tick represents 1 simulated minute. At this speed, 24 minutes of real time simulate one hospital day. The clock is the authoritative timekeeper — no module may advance or delay it. All random decisions use a seeded PRNG (mulberry32) initialized from the simulation seed, ensuring deterministic reproducibility.

Event journal. Every state transition is recorded in an append-only SQLite journal with WAL mode. Each entry captures the tick, timestamp, event type, entity, and a JSON payload. Full-state snapshots are serialized every 20 ticks, enabling replay from any checkpoint. The current state is a materialized view derived from the nearest snapshot plus subsequent journal entries, following the CQRS/event sourcing pattern.

Handler chain. Each tick processes state through a pipeline of 35 handler functions, each responsible for a specific domain (admission, outpatient, emergency, lab, pharmacy, nursing, doctor, radiology, surgery, blood bank, microbiology, pathology, CSSD, biomedical engineering, infection control, clinical nutrition, radiotherapy, dialysis, etc.). Handlers are pure functions that do not communicate directly — they share state only through the state object passed through the chain.

3.2 Key components

Patient generation and admission. A 50-diagnosis ICD-10 pool weighted for Tier A hospital admission patterns produces patient profiles with age-appropriate vitals, Indonesian identity data, blood type, and drug allergy probabilities. A Markov process evaluates bed availability, calendar event modifiers, and disaster surge multipliers every tick to determine admissions.

Rule-based clinical agents. Three agent types operate under a clinical constitution: the doctor rounds every 4 ticks, assigns specialists via 14-specialty ICD mapping, and generates clinical actions ranked by priority and historical effectiveness; the nurse monitors vitals and administers medications; the pharmacist reviews orders against drug allergies, contraindications, and dose ranges. Agents are rule-based, not learned, ensuring deterministic behaviour. The architecture supports swapping these for learned policies (RL, LLM) without changes to the handler chain.

Disaster scenario engine. Seven stochastic disaster types (earthquake, tsunami, pandemic, etc.) progress through four phases — ramping, sustained, recovering, resolved — affecting patient surge ($2-7\times$), mortality, supply demand, staff availability, and infrastructure.

Cultural calendar engine. Patient influx patterns are tied to Indonesian holidays: Ramadan admissions for dehydration and hypoglycemia, Lebaran surges for burn wounds from firecrackers and travel fractures, New Year for road trauma. No existing simulator models holiday-specific epidemiology.

FHIR R4 adapter. Internal simulation state is transformed into standard FHIR resources (Patient, Encounter, Observation) on demand, enabling external health information systems to consume simulated data as ground truth.

4. Illustrative Examples

In 10 seeded runs of 1000 ticks each (simulating ~16.7 hours of hospital operations per run), the platform produced consistent outcome distributions: average length of stay was 82.9 ticks (SD 10.4, 95% CI ± 6.4), maximum LOS reached 916-992 ticks confirming scheduled inpatient delays function correctly, and average deaths per run was 4.3 (SD 1.9, 95% CI ± 1.2). Bed occupancy averaged 130.9 of 133 beds (98%, SD 2.4). In runs where the stochastic engine triggered disasters (3 of 10), mortality was 82% higher (mean 6.0 vs 3.3 deaths).

These experiments demonstrate three properties: the platform produces deterministic trajectories from identical seeds; it sustains concurrent multi-department operations with 35 handlers per tick; and its FHIR adapter exposes standardized clinical data that external systems can consume.

5. Impact and Reusability

Deer's Rock supports four categories of use across different audiences:

Policy experimentation. Regulators can test counterfactuals — minimum ICU beds for tsunami scenarios, discharge policy effects on pandemic surge — against identical patient populations using seeded replay.

Health IT validation. The FHIR adapter enables a new testing paradigm: an EHR or clinical decision support tool connects to the simulation as if to a live hospital and validates against

verifiable ground truth, without requiring real patient data.

Agent benchmarking. Researchers can swap the rule-based agents for learned policies (RL, LLM) and compare against heuristic baselines on identical patient cohorts using the built-in outcome tracker.

Medical education. Snapshot replay enables reconstructing specific patient trajectories for teaching — an M&M case can be replayed from the tick where a critical decision was made.

Ethical considerations

The platform simulates Indonesian healthcare using cultural scenarios (Lebaran, Ramadan) that reflect real epidemiological patterns. These scenarios should be used with care: they are not stereotypes but evidence-based representations of how culture and health intersect. The mortality model assigns cause of death by terminal event, but 19% of deaths remain attributable to indirect causes — documented as a known limitation. All patient data is procedurally generated; no real patient records are used.

6. Conclusions

Deer's Rock is an open-source, persistent, event-driven hospital simulation that combines deterministic replay, modular composability, cultural contextualization, and health system interoperability in a single platform. The platform demonstration confirmed reproducible trajectories across multiple seeded runs, concurrent multi-department operation across 35 handlers, and FHIR-compliant data export.

The architecture is generalizable beyond hospitals: the handler chain pattern supports pharmaceutical supply chains, public health systems, and disaster response networks. Deer's Rock is available as open-source software, and all experiment data is published for independent verification.

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